COURSE INFORMATION

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School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSE 3203**	**Academic Session/Semester:**	**2026 2027 1**
Course name:	**Special Topics**	**Pre/co requisite (course name and code, if applicable):**	
Credit hours:	**3**		

Course synopsis	This course provides an intermediate-level study of current issues, technologies, and practices in software engineering, with a focus on Java web application development. It examines their application across the software development life cycle, including planning, analysis, design, implementation, testing, and maintenance. Topics include web application architecture, development environment configuration, Spring Web MVC, dynamic user interfaces, database integration, application programming interfaces (APIs), and application security. The course also addresses ethical considerations and responsible AI-assisted programming, including the critical review and validation of generated code. Through lectures, hands-on laboratories, problem-based activities, and a development project, students develop practical skills in implementation, collaboration, and critical problem-solving. Upon completion, students will be able to analyse software development problems, evaluate alternative technologies and solutions, design and implement Java web applications using appropriate software engineering practices.
Course coordinator (if applicable)	Dr Nor Azizah Binti Sa'adon

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Mr Norizam Katmon	N28		rizam@utm.my
	Dr Nor Azizah Binti Sa'adon	N28		azizahsaadon@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	[1]CLO	PLO [2](MQF Cluster Code)	[3]Taxonomies & [4]Graduate attributes	[5]ESD	[6]VBE	T&L methods	[7]Assessment methods
CLO1	Apply software engineering principles to solve given Java web application development problems, considering relationships among requirements, design, implementation, and maintenance.	PLO1 (C1)	C3 (SC1)			Active Learning,	Quiz: 5% Lab Exercise: 6% Test: 10% Final Exam: 12%
CLO2	Evaluate alternative designs for secure, database-driven Java web applications to justify an effective solution to a given problem using appropriate frameworks and techniques.	PLO2 (C2)	C5 (SC2)			Active Learning, Project-based Learning (PBL)	Quiz: 5% Lab Exercise: 9% Test: 10% Final Exam: 18% Assignment - Code: 5% Project Progress: 3% Project Code: 7%
CLO3	Demonstrate professional integrity and responsibility by confidently communicating the proposed alternative solution to an audience through an oral presentation, acknowledging limitations, and responding respectfully during Q&A.	PLO5 (C3C)	A3 (SI4)			Active Learning, PBL	Assignment - Video: 5% Project Final Presentation: 5%

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.

[1]**Up to 5 CLO**

[2]**MQF Cluster Code**

C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

[3]**Taxonomies of Learning Domain** and [4]**UTM's Graduate Attributes** for measurement of outcomes achievement

[5]**Education for Sustainable Development (ESD)**

Systems Thinking (ST); Anticipatory Thinking (AT); Normative (N); Strategic Thinking (STT); Collaboration (C); Critical Thinking (CT); Self-Awareness (SA; Integrated Problem Solving (IPS)

[7]**Value-Based Education (VBE)**

Integrity; Synergy; Excellence; Sustainability

[7]T – Test; Q – Quiz; Asg – Assignment; PR – Project; LE-Lab Exercise; F – Final Exam.

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active Learning	Conducted through in-class activities such as pair programming & group discussion
2	Problem-based Learning	Conducted through group project based on real-world problems, which tasks are given in sequential steps throughout the semester. Students in a group of 3-4 are asked to deliver a usable and tested interactive system interface.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

PLO5 (C3C) - Communicate effectively both in written and spoken form with engineers, other professionals and community.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1 12 Oct - 18 Oct	**Intro: Current Trends and Ethics** Java web evolution; HTTP overview; AI-assisted coding; responsible vibe coding; privacy; intellectual property; academic integrity; developer accountability.	CLO1	2	2							1	2	7
Week 2 19 Oct - 25 Oct	**Preparation: Development Setup** JDK; IDE-VScode; Maven; Tomcat; GitHub; creating, configuring and running a basic Java web project, CI/CD **Project Briefing and Team Formation**	CLO2 CLO3	1		3						1	2	7

Week 3 26 Oct - 1 Nov	**Management: SDLC and Project Planning** SDLC stages; Agile overview; project scope; team roles; task planning (JIRA); basic version-control workflow (Git), overview of software engineering principles.	CLO1 CLO3	1		3						1	2	7
Week 4 2 Nov - 8 Nov *(Deepavali 8 Nov)*	**Analysis: Requirements Analysis** User stories; use cases; functional requirements; business process; requirements validation; requirement traceability. **Lab Exercise 1**	CLO1	1	2	1						2	1	7
Week 5 9 Nov - 15 Nov (Deepavali 9 Nov)	**Design: Web Application Design** MVC; layered architecture; separation of concerns; modularity; basic UML class diagram; database design. **Lab Exercise 2**	CLO1	1	2	1						2	1	7
Week 6 16 Nov - 22 Nov	**Implementation: Spring Web MVC** Spring architecture; application context; DispatcherServlet; controllers; request mapping; Spring MVC request flow. **Lab Exercise 3**	CLO2	2	2							1	2	7
Week 7 23 Nov - 29 Nov	**Implementation: Web Views with Thymeleaf, Forms and Validation** Thymeleaf views; Spring Model; expressions; iteration; forms (input validation); @ModelAttribute; model binding; validation; error messages; redirect. **#Quiz 1**	CLO2	2	2							1	2	7
Week 8 30 Nov - 6 Dec	MID-SEM BREAK												
Week 9 7 Dec - 13 Dec	**Implementation: Dependency Injection and Layered Application Architecture** Inversion of control; constructor injection; interfaces; controller, service and repository layers; low coupling; high cohesion **Project Progress** **#Test - 10 December 2026 (Thursday) 8pm**	CLO2	2	1	1							3	7
Week 10 14 Dec 20 Dec	**Implementation: Persistence and Database Integration** Database connections; SQL operation; safe data access. **Assignment**	CLO2 CLO3	2	1	1							3	7
Week 11 21 Dec -27 Dec *(Christmas - 25 Dec)*	**Implementation: JPA and Hibernate** ORM concepts; entity mapping; identifiers; basic relationships; queries; CRUD operations. **Lab Exercise 4**	CLO2	2	2							3		7
Week 12 28 Dec - 3 Jan	**Implementation: RESTful Web Services and Fetch API** REST principles; @RestController; REST endpoints; HTTP methods; JSON; @RequestBody; status codes; ResponseEntity; Fetch API; basic error handling. **Lab Exercise 5**	CLO2	2	2							3		7

Week 13 4 Jan - 10 Jan	**Testing: Spring Security, Testing and Code Quality** Security by design; authentication; authorisation; BCrypt; RBAC; protected resources; CSRF awareness; JUnit; MockMvc; debugging; logging; code review; readable code; duplication reduction.	CLO2	2	2								3	7
Week 14 11 Jan -17 Jan	**Deployment and Maintenance** Executable JAR; Maven build; profiles; environment variables; logging; CI/CD basics; deployment; monitoring; maintenance; change-impact analysis **#Quiz 2**	CLO2	2	1	1						3		7
Week 15 18 Jan - 24 Jan *(Thaipusam - 22 Jan)*	**Evaluation: Project Demonstration** System demonstration; architecture explanation; test evidence; security review; AI-use declaration; technical reflection. **Project Final Presentation**	CLO3	1	1	2							3	7
											SUB-TOTAL SLT :		**98**

Continous Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Quiz 1 & 2	10	1				1
2	Assignment (Code + Video)	10			3	3	6
3	Lab Exercise (5 LE)	15	5				5
4	Group Project (Progress 3%, Presentation 5%, Code 7%)	15	2			3	5
5	Test	20	2				2
Assessment activities carried out during the semester to monitor student progress (e.g., quizzes, assignments, projects).					**SUB-TOTAL SLT :**		**19**

Final Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Final Exam	30	3				3
2							0
3							0
4							0
5							0
Assessment conducted at the end of the course to evaluate the achievement of overall learning outcomes (e.g., final exam, final project).					**SUB-TOTAL SLT :**		**3**
					SLT for Assessment:		**22**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	57.50
B	% SLT for Online & Independent Learning Component :	42.50
C	%SLT for Online Component:	17.50
D	% SLT for All Practical Component:	10.83
D1	% SLT for F2F Physical Practical Component:	10.83
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)
Visual Studio Code, Apache Maven, and Tomcat, Database - MySQL, Hibernate, and Spring Framework. Software management tools - Git, GitHub, Jira

References (include required and further readings, and should be the most current)
Main text: Spilca, L. (2021). Spring start here. Manning Publications. Additional references: 1. Spilca, L. (2024). Spring security in action (2nd ed.). Manning Publications. 2. Spring Team. (n.d.). Spring Framework reference documentation. Retrieved October, 2026, from https://docs.spring.io/spring-framework/reference/ Learning Management System (LMS) http://elearning.utm.my

Other additional information (if applicable)
Academic honesty and plagiarism:
Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

1-**Attendance:** Attendance is compulsory and will be recorded at every lecture session. Students whose attendance falls below 80% will not be permitted to sit for the final examination.
2-**Conduct and attire:** Students must observe the University's code of conduct, dress regulations, and standards of etiquette at all times.
3-**Class exercises and tutorials:** Exercises and tutorial activities will be assigned during class, and selected activities may contribute to coursework assessment. Students who do not complete assessed activities will receive no marks for those activities.
4-**Assignment submission:** Assignments must be submitted by the specified deadlines. Late submissions will incur mark deductions. Assignments submitted three or more days after the deadline will not be accepted.
5-**Make-up examinations:** Make-up examinations will be granted only to students who are medically unfit and provide a medical certificate verified by a UTM panel doctor. The make-up examination must take place within one week of the original examination date.

#IMPORTANT
All students must adhere to the following to pass a computing course:
i.Pass both Continuous Assessment AND Final Assessment.
ii.Get at least minimum pass marks (40%) and pass grade (D+) following UTM regulation.

The passing mark for:
i.Continuous Assessment is 30% of continuous marks.
ii.Final Assessment is 20% of the final marks.

If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark.

	Assessment	PLO 1	PLO 2	PLO5	TOTAL
		CLO1	CLO2	CLO3	
1	ASSIGNMENT		5	5	10
2	LAB EXERCISE (5)	6	9		15
3	QUIZ (2)	5	5		10
4	GROUP PROJECT		10	5	15
5	TEST (WRITTEN)	10	10		20
6	FINAL EXAM	12	18		30
TOTAL CLO		33	57	10	100
TOTAL PLO		33	57	10	100

Disclaimer:

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ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:	**Certified by:**
Name: Dr Nor Azizah Binti Sa'adon	**Name:** PM Dr Mohd Adham Isa
Signature:	**Signature:**
Date: 8-Oct-26	**Date:** 8 Oktober 2026